FORM TP 02025193

TEST CODE 02112020

MAY/JUNE 2025

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

CHEMISTRY

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
5. A data booklet is provided.
6. You may use a silent, non-programmable calculator to answer questions.
7. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
8. **If you use the extra page(s), you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Answer ALL questions.

MODULE 1: FUNDAMENTALS OF CHEMISTRY

1. (a) The sketch in Figure 1 below shows the first ionization energy of the eight elements of Period 3.

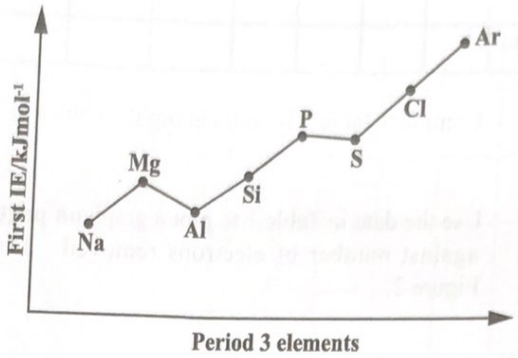

Figure 1. First ionization energy of the eight elements in Period 3

(i) State TWO factors which influence the values of the first ionization energy.

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[2 marks]

(ii) Explain how the ionization energy data provides evidence for the idea of subshells, using appropriate elements from the sketch in Figure 1.

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[4 marks]

GO ON TO THE NEXT PAGE

(b) Table 1 shows successive ionization energies for an element, D.

TABLE 1: IONIZATION ENERGIES (IE) FOR AN ELEMENT, D

No. of Electrons Removed	1	2	3	4	5	6	7	8	9	10	11	12
Ionisation Energy (IE) (kJ/mol)	690	1150	4910	6475	8146	10 500	12 320	14 207	18 194	20 491	57 152	63 433
Log_{10} IE	2.84											

(i) Complete Table 1 by calculating the values of Log_{10} IE (Ionization Energy).

[1 mark]

(ii) Use the data in Table 1 to plot a graph **on page 7**, of log_{10} IE (Ionization Energy) against number of electrons removed. The first point has been plotted in Figure 2.

[2 marks]

(iii) Explain the shape of the graph.

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[3 marks]

(iv) State the electronic configuration of Element D, based on the shape of the graph.

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[1 mark]

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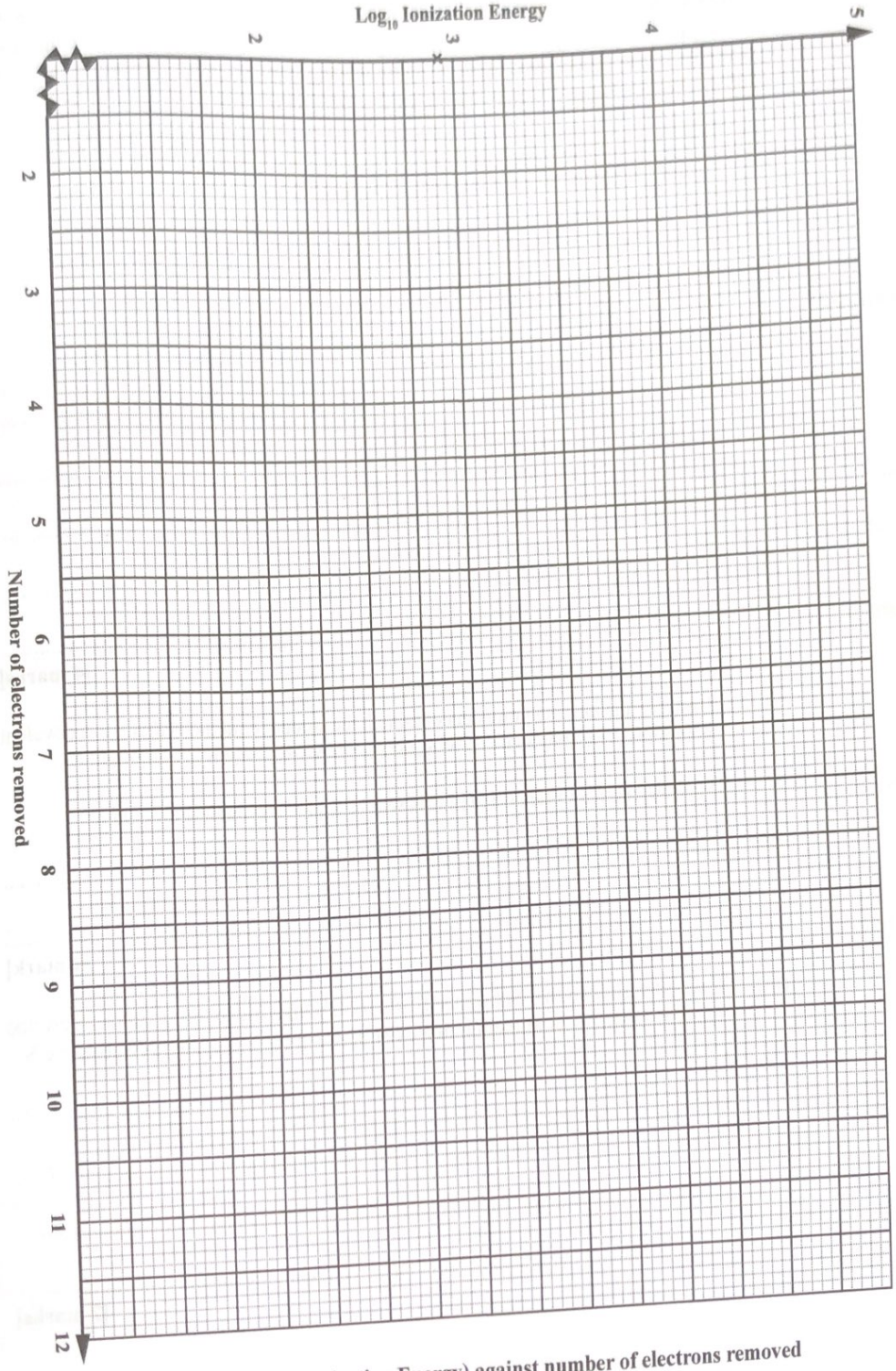

Figure 2. Log_{10} (Ionization Energy) against number of electrons removed

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(c) Substance S is a solid with a low melting point. A student is asked to determine the melting point of a sample of Substance S by using the following apparatus and materials available in the school laboratory: a thermometer, capillary tubes (sealed at one end), 500 mL beaker, Bunsen burner, tripod, gauze, tap water.

(i) Outline fully the steps involved in the procedure that the student would follow in order to obtain the melting point of Substance S.

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[4 marks]

(ii) Based on the melting point data obtained, the student concludes that S is a covalent compound. Further analysis of Substance S reveals that it is nonpolar.

State whether Substance S will dissolve in water or tetrachloromethane.

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[1 mark]

(iii) It was determined that Substance S is a solid with a low melting point. Describe the type of forces of attraction that exist between the molecules of Substance S.

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[2 marks]

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(d) Use the kinetic theory to explain EACH of the following.

(i) The change from a solid to a liquid state.

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[2 marks]

(ii) The nature of the liquid state.

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[2 marks]

(iii) The change from liquid to the gaseous state.

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[2 marks]

(e) A gas syringe contains 18.2 cm^3 of air at 55 °C. A volatile liquid with a mass of 0.185 grams is injected into the syringe. The volume of gas in the syringe then becomes 54.5 cm^3 at 55 °C and 1.01×10^5 Pa.

(i) Calculate the molar mass of the liquid. [R = 8.314 J K^{-1} mol^{-1}.]

[3 marks]

(ii) Suggest which measurement used in the calculation of the molar mass is subject to the greatest error.

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[1 mark]

Total 30 marks

MODULE 2: KINETICS AND EQUILIBRIA

2. (a) A mixture of hydrogen gas and iodine gas is sealed in a glass bulb and placed in a thermostat bath. A dynamic equilibrium is established at 450 °C.

(i) Describe the FOUR characteristic features of this dynamic equilibrium.

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[4 marks]

(ii) Write a balanced chemical equation to represent the reaction described in 2 (a) above.

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[2 marks]

(iii) It is more convenient to measure the pressure of gases at equilibrium rather than concentrations. Table 2 provides information on the analysis of the equilibrium mixture of the gases at 450 °C.

TABLE 2: EQUILIBRIUM PRESSURES OF THE SUBSTANCES IN THE MIXTURE

Substances	Hydrogen	Iodine	Hydrogen Iodide
Equilibrium Pressure/Pa	2.5×10^4	1.6×10^4	4.0×10^4

Calculate the K_p value of the equilibrium mixture.

[3 marks]

(b) (i) Describe EACH of the following terms.

Solubility product

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[2 marks]

Common ion effect

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[2 marks]

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(ii) Calcium phosphate, $Ca_3(PO_4)_2$ is a water-insoluble mineral which is used to make commercial fertilizers. Given that the concentrations of Ca^{2+} and PO_4^{3-} ions in equilibrium are 2×10^{-8} and 1×10^{-9} respectively, determine the solubility product, K_{sp}, of calcium phosphate.

[3 marks]

(iii) Write a balanced equation for the equilibrium reaction of the formation of calcium and phosphate ions from calcium phosphate.

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[2 marks]

(c) A saturated solution of silver chloride is filtered and the residue washed with dilute hydrochloric acid instead of water.

Comment on the solubility of silver chloride in water compared with the solubility of silver chloride in dilute hydrochloric acid and justify the difference, if any.

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[2 marks]

(d) Write the steps in an experimental procedure for the determination of the solubility product constant of barium hydroxide, $Ba(OH)_2$.

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[5 marks]

(e) Buffer solutions are used to provide suitable media for a variety of activities ranging from industrial to essential life processes.

(i) Define the term 'buffer solution'.

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[1 mark]

(ii) Explain how the molecular structure of amino acids relates to their function as buffer solutions in human blood.

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[3 marks]

(iii) State ONE industry in which buffer solutions are used.

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[1 mark]

Total 30 marks

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MODULE 3: CHEMISTRY OF THE ELEMENTS

3. (a) Table 3 shows the variation in some physical properties of Group IV elements.

TABLE 3: SOME PHYSICAL PROPERTIES OF GROUP 4 ELEMENTS

Elements	C(d)	Si	Ge	Sn	Pb
Melting point, °C	3550	1410	937	232	328
Electrical Conductivity ($ohm^{-1}m^{-1}$)	–	1×10^6	2×10^6	8×10^6	5×10^6
Melting point, of XO_2 X = Group IV element (°C)	–56	1610	1115	1630	290

C(d) = diamond

(i) Describe the trend in EACH of the following.

Electrical conductivity from silicon to lead

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[2 marks]

The melting point from silicon to lead

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[2 marks]

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(ii) Account for the variation in melting points from C(d) to Sn, in terms of structure and bonding.

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[4 marks]

(b) (i) Describe the reaction of lead (IV) chloride with water.

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[1 mark]

(ii) Write a balanced equation for the reaction described in (b) (i).

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[2 marks]

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(c) Table 4 shows some physical properties of the Group II elements.

TABLE 4: SOME PHYSICAL PROPERTIES OF THE GROUP II ELEMENTS

Element	Density (g/cm^3)	Atomic Radii (nm)	Ionic Radii (nm)
Be	1.85	0.122	0.031
Mg	1.74	0.160	0.065
Ca	1.55	0.197	0.099
Sr	2.60	0.215	0.113
Ba	3.51	0.224	0.135

(i) State the trend in EACH of the following from beryllium to barium.

Density

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[1 mark]

Atomic radii

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[1 mark]

(ii) Account for the trend in the atomic radii in terms of the nuclear charge and the screening effect.

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[3 marks]

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(d) (i) Complete the table below by recording the observations that would be made when a 1 molar (M) solution of sodium sulfate is added to 0.1 mol dm^{-3} of each of the Group II cations.

Group II Cation	Observations: Addition of 1 M Na_2SO_4 Solution
Mg^{2+}	
Ca^{2+}	
Sr^{2+}	
Ba^{2+}	

[4 marks]

(ii) Account for the observations in (d) (i).

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[4 marks]

(iii) Write the ionic equation for the reaction between the barium ion, Ba^{2+} and the 1 M sodium sulfate solution.

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[2 marks]

(e) Some compounds of aluminium, calcium and magnesium have various uses. For any TWO of the metals, complete the following table by naming ONE compound of the metal and the use of the compound.

Metal	Name of Metallic Compound	Use of Metallic Compound

[4 marks]

Total 30 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.